

EXERCISE SET 3.1*

3.1.33 Let $\mathbf{R}$ be the domain of the predicate variables a , b , c , and d . Which of the following are true and which are false? Give counterexamples for the statements that are false.

a. $a > 0$ and $b > 0 \rightarrow ab > 0$

b. $a < 0$ and $b < 0 \rightarrow ab < 0$

c. $ab = 0 \rightarrow a = 0$ or $b = 0$

d. $a < b$ and $c < d \rightarrow ac < bd$

$a, b, c, d \in \mathbf{R}$

a. $a > 0$ and $b > 0 \rightarrow ab > 0$

In English:

“If a is greater than 0 and b is greater than 0, then ab is greater than 0.”

Solution: The statement is true $\forall a, b \in \mathbf{R}$

b. $a < 0$ and $b < 0 \rightarrow ab < 0$

In English:

“If a is less than 0 and b is less than 0, then ab is less than 0.”

Solution: The statement is false, because if a is -1 and b is -2, then ab is 2, which's greater than 0.

c. $ab = 0 \rightarrow a = 0 \text{ or } b = 0$

In English:

“If ab is 0, then a is 0 or b is 0.”

Solution: The statement is true $\forall a, b \in \mathbf{R}$

d. $a < b \text{ and } c < d \rightarrow ac < bd$

In English:

“If a is less than b and c is less than d , then ac is less than bd .”

Solution: The statement is false, because if a is -0.1, and b is 0.11, and c is -10.1 and d is 1.11, then ac is 1.01 and bd is 0.1221. Therefore, ac is greater than bd .

Solutions: a. The statement is true $\forall a, b \in \mathbf{R}$

b. The statement is false, because if a is -1 and b is -2, then ab is 2, which's greater than 0.

c. The statement is true $\forall a, b \in \mathbf{R}$

d. The statement is false, because if a is -0.1, and b is 0.11, and c is -10.1 and d is 1.11, then ac is 1.01 and bd is 0.1221. Therefore, ac is greater than bd .